

SudokuHelper

Description

This small plugin helps with solving Sudoku puzzles.

However, as the name suggests, it is only a "helper", a utility program; it does not solve a Sudoku automatically.

Information

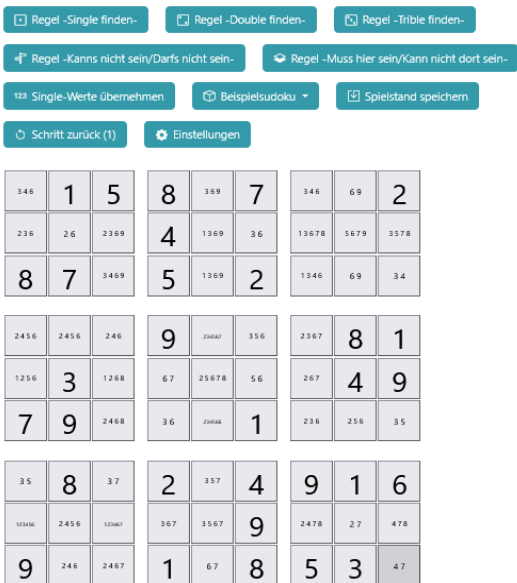
Author:	rmb
Plugin version:	3.2
Required Admidio version	5.0
Supported databases:	MySQL
Supported languages:	German (informal), German (formal), English
License:	GPL 2
Source code:	GitHub

Screenshots

SudokuHelper v3.2 Beta 1



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How it works

An empty Sudoku puzzle is structured as follows: It is a square consisting of 9 rows and 9 columns, for a total of 81 boxes. This square is further subdivided into sub-squares, each containing 9 boxes.

Each box can contain a number between 1 and 9. This is displayed as follows when SudokuHelper starts:



Each box is designed as a "button" and can be clicked.

For example, if you want to set a box to a specific value, you click on the box.

The following window will open:

setzen		möglich
☐	1	☒
☐	2	☒
☐	3	☒
☐	4	☒
☐	5	☒
☐	6	☒
☐	7	☒
☐	8	☒
☐	9	☒

Speichern

The column "possible" ("möglich") indicates that all numbers from 1 to 9 would still be possible for this box.

The "set" ("setzen") column allows you to set a box to a specific number.

In the following example, row 5, column 6 was set to "3":

123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789

As a result, the “3” cannot be placed in any other box of row 5, column 6 and the sub-square in which the “3” is located.

All boxes that were changed in this step are highlighted in color for better visibility.

Another example:

3 4 6	1	5	8	3 6 9	7	3 4 6	6 9	2
2 3 6	2 6	2 3 6 9	4	1 3 6 9	3 6	1 3 6 7 8	5 6 7 9	3 5 7 8
8	7	3 4 6 9	5	1 3 6 9	2	1 3 4 6	6 9	3 4
2 4 5 6	2 4 5 6	2 4 6	9	2 3 6 4 7	3 5 6	2 3 6 7	8	1
1 2 5 6	3	1 2 6 8	6 7	2 5 6 7 8	5 6	2 6 7	4	9
7	9	2 4 6 8	3 6	2 3 6 4 8	1	2 3 6	2 5 6	3 5
3 5	8	3 7	2	3 5 7	4	9	1	6
1 2 3 4 5 6	2 4 5 6	1 2 3 4 6 7	3 6 7	3 5 6 7	9	2 4 7 8	2 7	4 7 8
9	2 4 6	2 4 6 7	1	6 7	8	5	3	4 7

In row 6, the combination 4 & 8 is only present in column 3 and column 5.

To indicate this, remove all impossible numbers from these two boxes. To do this, click on the box in row 6, column 3 and remove the impossible numbers 2 and 6.

SudokuHelper

setzen		möglich
☐	1	☐
☐	2	☒
☐	3	☐
☐	4	☒
☐	5	☐
☐	6	☒
☐	7	☐
☐	8	☒
☐	9	☐

✓ Speichern

SudokuHelper

setzen		möglich
☐	1	☐
☐	2	☐
☐	3	☐
☐	4	☒
☐	5	☐
☐	6	☐
☐	7	☐
☐	8	☒
☐	9	☐

✓ Speichern

Result:

3 4 6	1	5	8	3 6 9	7	3 4 6	6 9	2
2 3 6	2 6	2 3 6 9	4	1 3 6 9	3 6	1 3 6 7 8	5 6 7 9	3 5 7 8
8	7	3 4 6 9	5	1 3 6 9	2	1 3 4 6	6 9	3 4

2 4 5 6	2 4 5 6	2 4 6	9	2 3 6 7	3 5 6	2 3 6 7	8	1
1 2 5 6	3	1 2 5 8	6 7	2 5 6 7 8	5 6	2 6 7	4	9
7	9	4 8	3 6	2 3 6 7 8	1	2 3 6	2 5 6	3 5

3 5	8	3 7	2	3 5 7	4	9	1	6
1 2 3 4 5 6	2 4 5 6	1 2 3 6 7	3 6 7	3 5 6 7	9	2 4 7 8	2 7	4 7 8
9	2 4 6	2 4 6 7	1	6 7	8	5	3	4 7

Proceed in the same way with the box in row 6, column 5.

SudokuHelper ✕

setzen		möglich
☐	1	☐
☐	2	☒
☐	3	☒
☐	4	☒
☐	5	☒
☐	6	☒
☐	7	☐
☐	8	☒
☐	9	☐

✓ Speichern

SudokuHelper ✕

setzen		möglich
☐	1	☐
☐	2	☐
☐	3	☐
☐	4	☒
☐	5	☐
☐	6	☐
☐	7	☐
☐	8	☒
☐	9	☐

✓ Speichern

Result:

3 4 6	1	5	8	3 6 9	7	3 4 6	6 9	2
2 3 6	2 6	2 3 6 9	4	1 3 6 9	3 6	1 3 6 7 8	5 6 7 9	3 5 7 8
8	7	3 4 6 9	5	1 3 6 9	2	1 3 4 6	6 9	3 4
2 4 5 6	2 4 5 6	2 4 6	9	2 3 4 5 6 7	3 5 6	2 3 6 7	8	1
1 2 5 6	3	1 2 6 8	6 7	2 5 6 7 8	5 6	2 6 7	4	9
7	9	4 8	3 6	4 8	1	2 3 6	2 5 6	3 5
3 5	8	3 7	2	3 5 7	4	9	1	6
1 2 3 4 5 6	2 4 5 6	1 2 3 4 6 7	3 6 7	3 5 6 7	9	2 4 7 8	2 7	4 7 8
9	2 4 6	2 4 6 7	1	6 7	8	5	3	4 7

Sudoku can be solved step by step by placing a box on a number and removing impossible numbers.

To assist with solving puzzles, SudokuHelper includes several predefined rules (or strategies). These are described in the section "The Menu Bar".

The installation

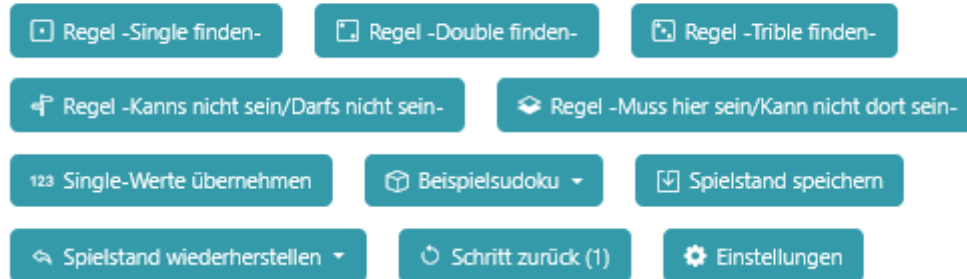
The following steps must be performed for installation:

1. Place all plugin files in the folder
<your Admidio installation>/adm_plugins/ SudokuHelper . (If the folder does not yet exist, it must be created).
2. Have an administrator start the installation script.
To do this, open the following file:
<your Admidio installation> /adm_plugins/SudokuHelper/system/install.php

The installation routine creates a menu item with access rights.

The menu bar

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The Rules

The “rules” described below are the result of my years of experience with Sudoku. They are not official Sudoku rules, but rather insights gained from experience—or my own personal strategies (rmb).

The button “Rule: -Find single numbers-”

This rule finds numbers that occur only once.

- in one line
- in a column
- or in a sub-square

Example:

12468	23689	23459	23459	258	23689	2578	24567	24678
248	2589	7	2459	6	1	3	245	248
2468	23568	2345	2345	258	7	258	9	1
27	4	8	123	9	23	6	127	5
9	2567	25	1256	12578	2568	127	1247	3
3	2567	1	256	4	256	9	8	27
5	1	2349	8	2	2469	27	2367	2679
28	289	6	7	3	259	4	125	289
2478	23789	2349	12689	125	24569	12578	12347	26789

In the marked sub-square, the numbers 7 and 8 each appear only once.

The rule "Find single numbers " identifies these "singles" and removes all other possible numbers.

1	2345678	23459	23459	258	2345678	2578	24567	24678
248	2589	7	2459	6	1	3	245	248
6	23568	2345	2345	258	7	258	9	1

27	4	8	123	9	23	6	127	5
9	2567	25	1256	7	8	127	4	3
3	2567	1	256	4	256	9	8	27

5	1	2349	8	2	2469	27	2367	2679
28	289	6	7	3	259	4	1	289
2478	23789	2349	12345678	125	24569	12578	12367	26789

These "single values" can be directly copied and saved using the "Accept single values" button.

1	23589	23459	23459	58	23459	258	2567	24678
248	2589	7	2459	6	1	3	25	248
6	2358	2345	2345	58	7	258	9	1

27	4	8	123	9	23	6	27	5
9	256	25	1256	7	8	12	4	3
3	2567	1	256	4	256	9	8	27

5	1	349	8	2	469	7	36	69
28	289	6	7	3	59	4	1	289
2478	23789	2349	14569	15	4569	258	2356	2689

The button "Rule: -Find double numbers-"

This rule applies to pairs of two,

- in one line
- in a column
- or in a sub-square

and removes all other possible numbers.

Example:

3 4 6	1	5	8	3 6 9	7	3 4 6	6 9	2
2 3 6	2 6	2 3 6 9	4	1 3 6 9	3 6	1 3 6 7 8	5 6 7 9	3 5 7 8
8	7	3 4 6 9	5	1 3 6 9	2	1 3 4 6	6 9	3 4
2 4 5 6	2 4 5 6	2 4 6	9	2 3 4 5 6 7	3 5 6	2 3 6 7	8	1
1 2 5 6	3	1 2 6 8	6 7	2 5 6 7 8	5 6	2 6 7	4	9
7	9	2 4 6 8	3 6	2 3 4 5 6 8	1	2 3 6	2 5 6	3 5
3 5	8	3 7	2	3 5 7	4	9	1	6
1 2 3 4 5 6	2 4 5 6	1 2 3 4 6 7	3 6 7	3 5 6 7	9	2 4 7 8	2 7	4 7 8
9	2 4 6	2 4 6 7	1	6 7	8	5	3	4 7

In the row marked in red, the combination 4,8 appears exactly twice.

The "Find Double" rule recognizes this combination and removes all other possible numbers.

3 4 6	1	5	8	3 6 9	7	3 4 6	6 9	2
2 3 6	2 6	2 3 6 9	4	1 3 6 9	3 6	1 3 6 7 8	5 6 7 9	3 5 7 8
8	7	3 4 6 9	5	1 3 6 9	2	1 3 4 6	6 9	3 4
2 4 5 6	2 4 5 6	2 4 6	9	2 3 4 5 6 7	3 5 6	2 3 6 7	8	1
1 2 5 6	3	1 2 6 8	6 7	2 5 6 7 8	5 6	2 6 7	4	9
7	9	4 8	3 6	4 8	1	2 3 6	2 5 6	3 5
3 5	8	3 7	2	3 5 7	4	9	1	6
1 2 3 4 5 6	2 4 5 6	1 2 3 4 6 7	3 6 7	3 5 6 7	9	2 4 7 8	2 7	4 7 8
9	2 4 6	2 4 6 7	1	6 7	8	5	3	4 7

The button “Rule: -Find trible numbers-”

This rule applies to pairs of three,

- in one line
- in a column
- or in a sub-square

and removes all other possible numbers.

The button “Rule: -Cannot be/Must not be-”

Rule: If a number cannot be in one area, it cannot be in another either.

Example:

12468	23569	23459	23459	258	23568	2578	24567	24678
248	2589	7	2459	6	1	3	245	248
2468	23568	2345	2345	258	7	258	9	1
27	4	8	123	9	23	6	127	5
9	2567	25	1256	12578	2568	127	1247	3
3	2567	1	256	4	256	9	8	27
5	1	2349	8	2	2469	27	2367	2679
28	289	6	7	3	259	4	125	289
2478	23789	2349	124569	125	24569	12578	123567	26789

12468	23569	23459	23459	258	23568	2578	24567	24678
248	2589	7	2459	6	1	3	245	248
2468	23568	2345	2345	258	7	258	9	1
27	4	8	123	9	23	6	127	5
9	2567	25	1256	12578	2568	127	1247	3
3	2567	1	256	4	256	9	8	27
5	1	2349	8	2	2469	27	2367	2679
28	289	6	7	3	259	4	125	289
2478	23789	2349	124569	125	24569	12578	123567	26789

The number 4 cannot be in the blue zone and therefore must not be in the green zone either.

Result:

1248	23589	23459	23459	258	23459	2578	2567	24678
248	2589	7	2459	6	1	3	25	248
2468	2358	2345	2345	258	7	258	9	1
27	4	8	123	9	23	6	127	5
9	2567	25	1256	12578	2568	127	1247	3
3	2567	1	256	4	256	9	8	27
5	1	2349	8	2	2469	27	2367	2679
28	289	6	7	3	259	4	125	289
2478	23789	2349	136549	125	24569	258	2356	2689

The button “Rule -Must be here/Cannot be there-”

Rule: Certain numbers must be here and therefore cannot be there.

Example:

123456789	123456789	123456789	235689	235689	235689	134567	134567	134567
356	356	356	4	7	1	8	2	9
123456789	123456789	123456789	235689	235689	235689	134567	134567	134567

In this line, in the green area, the numbers 3, 5 and 6 still need to be inserted.

Therefore, they cannot occur in the areas marked in red.

123456789	123456789	123456789	235689	235689	235689	134567	134567	134567
356	356	356	4	7	1	8	2	9
123456789	123456789	123456789	235689	235689	235689	134567	134567	134567

Result after applying the rule:

124789	124789	124789	235689	235689	235689	134567	134567	134567
3 5 6	3 5 6	3 5 6	4	7	1	8	2	9
124789	124789	124789	235689	235689	235689	134567	134567	134567

The button “Rule –Clean up pairs–”

This rule identifies identical combinations

- in one line
- in a column
- or in a sub-square

that have arisen during the normal solving process, and clears the corresponding row, column, or sub-square.

Example:

123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
234568	123456789	2 5	123456789	123456789	2 5	123456789	123456789	123456789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789

In this row, two cells contain 2 and 5 as possible values. The 2 and the 5 can only be placed in these two cells, and not in any other cell in this row. Consequently, the 2 and the 5 cannot appear in the remaining cells of this row.

Result after applying the rule:

123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789
3 4 6 8	1346789	2 5	1346789	1346789	2 5	1346789	1346789	1346789
123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789	123456789

The button "Save single values"

The "Accept single values" button allows you to accept and save individual numbers.

The button "Sample Sudoku"

Two example Sudoku puzzles can be displayed here for testing.

The button "Save the game"

Saves the current game progress with date and time.

The button "Restore the game"

Restoring a saved state

The button "Step back"

Every change in the Sudoku puzzle is saved in a history. You can go back one step in the history at a time.

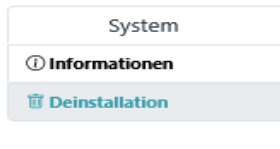
The button "Preferences"

The "Preferences" menu

The "Preferences" menu is divided into two menu items:

- Information
- Uninstallation

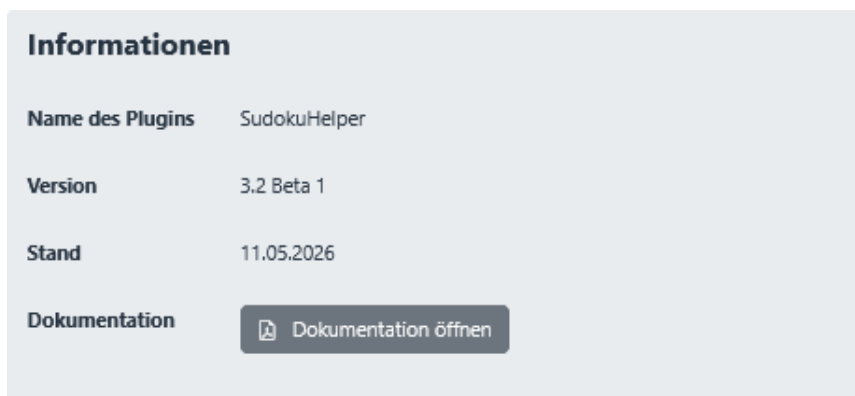
Einstellungen



Menu item “Information”

General information about the plugin, such as version and status.

The documentation can also be accessed here.



Menu item “Uninstallation”

The plugin can be removed via the "Uninstallation" option.

